

programming script. Based on their capabilities, the robots themselves split the main goal or task into smaller tasks, hence 'distributed.' Currently, COMRADE employs only two robots, Adam and Eve.

For example, a robot without a gripper assigns itself tasks that do not need one. Each robot possesses a highly layered architecture; each layer being capable of interaction at various levels.

Though it currently involves just two autonomous robots, the system is expansible to control any number of them. Adam and Eve are "differentially steered vehicles." Eve is equipped with an "onboard robotic gripper with three degrees of freedom."

Both have computers onboard for robot-logic control and image processing. Adam runs on a 2.4 GHz Athlon processor. Eve, as of now, is controlled by a laptop with a 115 kbps serial link attached to a Simputer.

An important innovation was using WiFi for communication between the robots. An 802.11b version of the WiFi link connects the robots. But why WiFi? Mrinal explains, "With a PC onboard, WiFi was easier to install, and significantly cheaper. You just needed to add a LAN card!"

"Nice weather, 50 million dollars, and no fishing boats—these are all the requirements for launching a craft to Mars"

Das, Mrinal, and Sharad Nagappa were classmates during the Telecommunication Engineering course they took at PESIT. Abhishek Sen Gupta was the odd man out, studying Electronics and Communications Engineering. Das or JD is from Orissa, Sen Gupta from Kolkata, and Mrinal and Sharad are from Bangalore.



Eve gripping a box while Adam looks on—a revolution at work

Little did the four know that their robots assembled at a makeshift workshop—an erstwhile kitchen—would prove that the "50 million dollars" element is not true.

The cumulative cost, as their accounts show, is about Rs 50,000. Of that, around Rs 16,000-20,000 came from sponsors—Yahoo! Inc and Encore Software. Samyeer Metrani of Encore also promised them a Simputer with a WiFi slot. The innovation that went into COMRADE is revolutionary, in that it used the most unlikely components to achieve uncharacteristically sophisticated results.

Mrinal, well-known in Bangalore for his involvement with the Bangalore Linux community, and his way with computer networks, dove into robotics by the end of the second year when he started working on ROAMER, a precursor to COMRADE. He was the driving force behind all



"We used antenna rods, broken CD-ROM drives, tape recorders and torch-lights!"

Jnaneshwar Das

Tech Specs

The Vehicle

The vehicle chassis is an aluminium framework that can negotiate through rough terrain, house the robotic arm and carry payloads when required. The onboard power source is a 12V, 32Ah car-battery which can power the vehicle for at least an hour. The 'robotics fabrication lab' was a converted kitchen in Jnaneshwar Das's house. Both vehicles have a maximum speed of 50 cm/s.

Software And Control

A microcontroller from the AVR series provides the low-level control. The microcontroller handles actuator control, monitoring of feedback data from encoders and the triggering and control of the SONAR range finders.

A Simputer supplies control information for the microcontroller board via a serial link. It uses the StrongARM SA-1110 processor whose speed is equivalent to that of a 400 MHz Intel Celeron. The Simputer handles image processing and obstacle avoidance, and the onboard PC controls WiFi communication with other robots. Encore Software has promised to give a Simputer with a WiFi slot in the near future.

Sensors

The sensors used on ADAM are SONAR range-finders, Web cams and electronic compasses. The range-finders map the environment and data is processed by the computer. Image data from the cameras are processed by IRIS.



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Buzzword
of the MONTH

Biometrics

Biometrics is determining identity based on physiology: using fingerprints, iris-

recognition, hand geometry, and facial recognition. Microsoft has unveiled its range of products with biometric security features. These

include mice and keyboards with built-in fingerprint recognition capabilities which will replace your password screen and protect your computer!